

Quality Control Through Statistical Methods

Quality control is important to minimise rejections during production of a proven design. Minimum number of rejections or increase in first-time yield saves production costs, testing time and resources. It also helps in reducing the cost of an item and increase customer satisfaction

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Quality control not only helps to inspect a product and give an acceptance/rejection decision, but also helps in reducing the number of rejections over time. For this, we need to analyse quality data using statistical methods and get required information on present quality. The information obtained should be used judiciously to improve the quality of a product. In this way, the decisions will not be biased by individual perceptions.

Total Quality Management (TQM) and Six Sigma processes to improve product quality are based on statistical methods. This article discusses some of the statistical methods used in quality control work. Emphasis is given on application and methodology rather than mathematics and related proofs.

The most popular statistical method amongst engineers is sampling technique. Other statistical methods include estimation of limits of a qual-

ity parameter with known confidence, statistical control charts, comparing quality parameters of two different lots of units (could be from different vendors), fitting quality data to a theoretical probability distribution function and proportion within specification limits.

Sampling

Sampling is used to accept/reject a large quantity of goods by inspecting a small sample from it. Certain small quantity (as per sampling plan) is selected at random from the goods. Every piece of the sample has to pass or fail a test as per definition given in the test procedure. If failed number of units is less than a defined acceptance number, the entire lot is accepted. Else the entire lot is rejected.

Since acceptance/rejection of the goods is dependent on testing of only a few samples from the lot, risks are involved in this process. Sometimes, a good lot can get rejected and a bad lot can get accepted. The probability of rejecting a good lot is called producer's

risk. The probability of accepting a bad lot is called consumer's risk. Graph of probability of acceptance versus percentage of defectives in a lot is shown in Fig. 1. It is called operating characteristic.

It can be observed that probability of acceptance decreases as percentage of defectives in a lot increases. Hence a supplier should minimise percentage of defectives in a lot so that the lot could be accepted with high probability. The supplier and consumer should agree on risks acceptable to both of them.

Confidence interval

Many a times, it is necessary to estimate properties of goods from a sample data. For example, if a random sample of 20 capacitors is taken from a large lot of 1000 μF bin, and their average is found to be 995 μF , the sample average 995 μF is called point estimate of lot average. This number can be used to estimate population (product) average with certain confidence, which is called confidence interval.

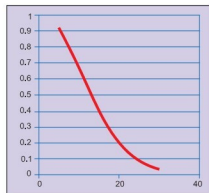


Fig. 1: Operating characteristic curve for a sampling scheme; Y-axis - P_a = Probability of lot acceptance; X-axis - Percentage of defectives in a lot

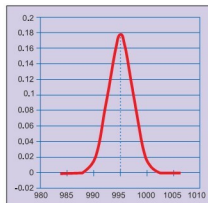


Fig. 2: Graph of normal distribution with 995 μF population average

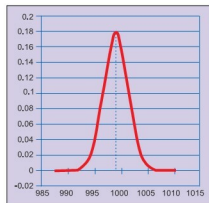


Fig. 3: Graph of normal distribution with 999 μF population average